

Jigsaw: All Together Practice!

Name/Write formulas for the following compounds.

Formula	Type of bond (ionic/covalent)	Name
Fe_2S_3	ionic	Iron (III) Sulfide
$\text{AlPO}_4 \cdot 7\text{H}_2\text{O}$	ionic	aluminum phosphate heptahydrate
Si_2Br_6	covalent	disilicon hexabromide
NaBr	ionic	sodium bromide
C_3H_8	covalent	tricarbon octahydride (propane)
Na_2SO_4	ionic	sodium sulfate
$\text{Co}_3(\text{PO}_4)_2$	ionic	Cobalt (II) phosphate
CF_4	covalent	Carbon tetrafluoride
$\text{Ca}(\text{NO}_2)_2$	ionic	Calcium nitrite
N_2O_3	covalent	dinitrogen trioxide
VO_2	ionic	Vanadium (IV) oxide
$\text{LiBr} \cdot 4\text{H}_2\text{O}$	covalent ionic	lithium bromide tetrahydrate
SrO	ionic	Strontium oxide
Ag_3P	ionic	Silver phosphide
H_2SO_4	ionic	Sulfuric acid
Cd_3N_2	ionic	cadmium (II) nitride
P_4Se_3	covalent	tetraphosphorus triselenide
$\text{Cu}_3(\text{PO}_4)_4 \cdot 5\text{H}_2\text{O}$	ionic	copper (IV) phosphate pentahydrate
ZnO	ionic	zinc oxide
H_3PO_4	covalent	hydrophosphoric acid
AlCl_3	ionic	aluminum chloride
K_2CO_3	ionic	potassium carbonate
HBrO_4	covalent ionic	perbromic acid
N_2O_3	covalent	dinitrogen trioxide
Ni_2S_3	ionic	nickel (III) sulfide
$\text{Sn}(\text{ClO}_3)_4$	ionic	tin (IV) chlorate
IF_5	covalent	iodine pentafluoride
$(\text{NH}_4)_2\text{O}$	ionic	ammonium oxide
BF_3	covalent	boron trifluoride

1. A halide is a binary compound, of which one part is a halogen atom and the other part is an element or radical that is less electronegative than the halogen, to make a fluoride, chloride, bromide, iodide or astatide compound. What is the general formula for alkali metal halides? $\rightarrow$ all M^{+1} X^{-1}

a. MX

b. M_2X ~~c. MX_2~~ d. MX_3

2. Which electrons account for many of the chemical and physical properties of elements?

a. Innermost

b. Intermediate

c. Outermost

d. Transition

(valence)

3. Which of the following elements is not correctly paired with its group (family) name?

a. Bismuth (Bi), halogens

c. Lithium (Li), alkali metals

b. Strontium (Sr), alkaline earth metals

d. Radon (Rn), noble gases

4. The inertness of noble gases is due to:

a. The number and arrangement of their electrons

b. The unique structure of their nuclei

c. The special number of protons and neutrons

d. The bonds they form with other elements

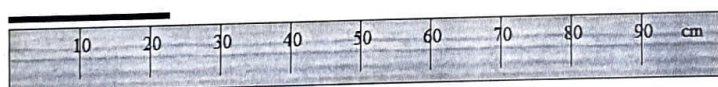
 $\rightarrow$ full outermost shell!

Sig Fig Practice

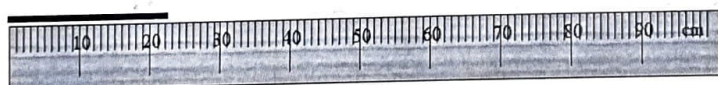
Directions: Measure the following to the correct number of significant figures.



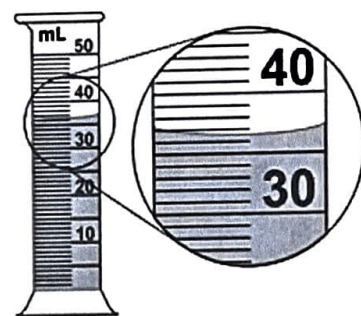
1. 20 cm



2. 22 cm



3. 22.2 cm



4. 36.5 mL

Directions: Determine the number of significant figures.

1. 0.02 12. 142 33. 0.02020 44. 0.073 25. 501 36. 1.071 47. 501.0 48. 10810 49. 5000 110. 5.00 311. 5000. 412. 1.20×10^3 3

Directions: Round your answer to the correct number of sig figs.

	Calculator Answer	Rounded Answer (with Correct # of Sig Figs)
1. $140 \text{ cm} \times 35 \text{ cm}$	4900	4900 cm^2 $4.9 \times 10^3 \text{ cm}^2$
2. $0.003 \text{ L} + 0.0048 \text{ L} + 0.100 \text{ L}$	0.1078	0.108 L
3. $67.35 \text{ cm}^2 \div (1.401 \text{ cm} - 0.399 \text{ cm})$	67.21557	67.2 cm
4. $(6.23 \text{ cm} + 3.111 \text{ cm} - 0.05 \text{ cm}) \times 14.99 \text{ cm}$	139.272	139 cm^2
5. $3.14159 \text{ m} \times (4.17 \text{ m} + 2.150 \text{ m})$	19.855	19.9 m^2

Coulomb's Law

a) $\oplus$ $\ominus$

b) $\oplus$ $\ominus$

c) $\oplus$

$\ominus$

d) $\ominus$ $\ominus$



1. Which set of particles shown the left will experience:

- a. the greatest attraction to each other? b
b. the greatest repulsion from each other? d

2. The particles in (a) will experience less attraction to each other than the particles in (c) because:

- a. greater, the distance between them is less.
b. smaller, the distance between them is ~~less~~ greater!
c. the same, distance is irrelevant to force of attraction.

3. The particles shown above in (a) will experience less attraction to each other than the particles in (b) because:

- a. greater, the nucleus has one proton instead of two.
b. smaller, the nucleus has one proton instead of two.
c. the same, charge is irrelevant to force of attraction.

an electron in the lowest energy level would be most attracted to the nucleus of which element?

- a. lithium
b. sodium

c. potassium

d. rubidium

most protons!

5. A proton would be least repulsed by the nucleus of which element?

a. helium

b. neon

c. argon

d. krypton

least protons

Isotopes! Complete the chart below.

Hyphen notation	Isotope Notation	Mass #	Atomic #	Protons	Neutrons	Electrons
boron-10	$^{10}_5\text{B}$	5	5	5	5	5
phosphorus-31	$^{31}_{15}\text{P}$	31	15	15	16	15
zinc-66	$^{66}_{30}\text{Zn}$	66	30	30	36	30
cobalt-59	$^{59}_{27}\text{Co}$	59	27	27	32	27

Types of Matter

Directions: Identify the following as: element (E), compound (C), heterogeneous mix (He) or homogeneous mix (Ho).

a. Table salt C

b. Nitric acid (HNO_3) C

c. Sugar (Glucose) C

d. Carbon dioxide (CO_2) C

e. Milk Ho

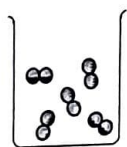
f. Air Ho

g. Nitrogen gas (N_2) E

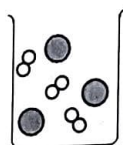
h. Zinc (Zn) E

i. Pulpy orange juice He

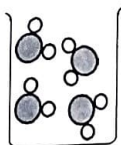
Directions: Use the particle representations below to answer #1-5. Answer choices may be used more than once!



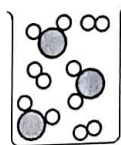
A



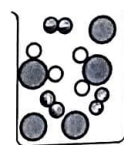
B



C



D



E

1. Compound C

2. Nitrogen, N_2 A

3. Mixture of two elements B

4. Element A

5. Mixture of water, H_2O , and hydrogen H_2 D

Directions: Choose the best possible answer for each question.

1. A forensic technician is examining a fine white powder found at a crime scene. It appears to be of uniform texture and consistency. He finds that upon heating, the substance decomposes, releasing a colorless gas and leaving a black residue. Based on these observations, the substance is:
A a homogeneous mixture C either a homogeneous mixture or a compound
B a heterogeneous mixture D a compound
2. An example of a heterogeneous mixture is
A bronze. B concrete. C brass. D water.
3. How is a mixture different from a compound?
A Particles of a mixture are combined chemically.
B Components of a mixture can only be separated chemically.
C Components of a mixture can be separated by physical means.
D Composition of a mixture may be constant.
4. A transparent liquid has uniform color. Under a microscope, no difference in uniformity is apparent. The liquid all boils away at the same temperature. Based on this information, it can be concluded that the liquid is:
A an element D either a pure substance or a homogeneous mixture
B a homogeneous mixture D a pure substance